UNIT-3(Image Restoration)

Image Degradation

Lecture-1

Image restoration is the operation of taking a corrupt/noisy image and estimating the clean, original image. Corruption may come in many forms such as motion blur, noise and camera mis-focus.



With the increasing demand for old classics back in fashion, such as old music, old clothing styles then why not old Photos? Taking this idea forward Image Restoration introduces ways to restore your old memories from the past with a touch of modern editing techniques. Every problem needs to be fixed whether it is in your life or in your photos!

Image Restoration Service is an appreciable service to recover your digital photos and digital assets. Numerous and varied functions can redefine your experiences and make them free from any sort of deterioration. Many factors such as age, water, and dust can make images dull and drab over the years like jewelry. With Photo Restoration Service amazing quality applications enable all damaged areas to be restored providing the reflection that it was never damaged.

With the advent of many new technologies, ways and methods to restore images close to reality are pretty much easier likewise in 3D. Such images give you a glimpse of the moment with clarity and sensation. Whether you want to remove scratches from your old photos online, mask them or clip them around the edges. Image Restoration Service is an all-rounder to convert your image into a sensational photograph.

What is Image Restoration?

Image Restoration is taking out the corrupt and noisy image and placing well-formed image. Its literal meaning implies "Restoring the Imagery". By descriptive meaning, we find it a helpful discipline originated from photo manipulation to bring back the lost vibe of photos. Corruption can be of many types such as a blur, a smudge, a destroyed edge, and camera misfocus. The process of photo restoration is performed by reversing the blurred process of the image by pointing out the point source in the image. It is called the Point Spread Function.

Image Restoration is not to be intermingled with Image Enhancement. These are two different terms and are out for different image objectives. The former works specifically to make the image more real and subtle, while the latter is significantly used to enhance the image in its glory.

Categories of Image Restoration

Image Restoration holds many varieties to choose from according to your image condition. Such as Portrait Restoration, Digital Photo Enhancement, Photographic reproduction of old images, Water Damaged Photo Restoration, Light Damaged Photo Restoration, Antique Image Restoration, and many more. All such categories prove helpful in eliminating the deepest level of imperfection and inaccuracy. Such editing functions can make your image as new as the moment it was clicked!

Benefits of Photo Restoration Service

A Professional Image Restoration Service holds many clear benefits that provide you with clean, clear, and originally restored images for your business advantage. All amazingly rich benefits are listed below for your convenience, Such as:

Preserving Originals

Image Restoration Service is admirable to restore your old yet important images to perfection. It can completely transform all your damaged, manipulated and water corroded images in a manner like the damage were never done. We understand the importance of your precious memories and go the extra mile to restore images the right way leaving no room for blemishes, imperfection, or the slightest smudge.

Eliminate Scratches, Folds, Tears, and Fading

Elimination of scratches from images is not as easy as it might seem. Obtaining picture-perfect results can be achieved only by the supervision of a professional. Photo Manipulators that can

cut short and remove all unwanted freckles, folds, watermarks, faded edges, and scratches for better resolution by increasing Photo Pixel Range.

Different Varieties To Choose From

Image Restoration Service is most commonly referred to as old photos. Old images require much hard work than today's advanced pixel images. There are tons of new-fashioned varieties that go excellent in restoring old photos. By photographic representation and reproduction of old photos copies of respective images are made. In such a way, old photos can be copied and all deformation can be cleaned properly in an orderly manner.

Like, all scratches can be filled, Missing parts and cracks can be reconstructed and enlarged accordingly. Pixel Magnification and Resolution can overall fancy your images, making them more prominent and kill extra blur. Copies of images can be used to identify and correct different red points at once. ITS Team can fully optimize your old images at your given time.

Electronic Storage

Professional service helps you convert images from different formats including PNG, GIF, PSD, JPEG, and TIFF, etc. Images can not only be restored but also accessible in any desirable format. They enable digitally accessible photos in portable formats for faster and easy transmission. All images can be transformed accordingly with the nature of the Business Company or Individual demand for effective and timely communication.

Marketing Benefits

There are numerous marketing benefits that you get by outsourcing bulk image restoration projects to a Professional Image Restoration Service Provider. ITS Team is proficient in getting your product images or old family photos processed and fixed. Photos can be reproduced from slides, negatives including old glass and positives, etc.

Techniques for Restoring Images

Images can become outdated due to adverse effects of light, pets, and children as well as many other complications. In Image Restoration, A mix and match of different techniques s the right way to achieve realistic results. Here are the two most popular Image Restoration Techniques employed by Professional Image Restoration Experts.

Photo Masking Technique

The most traditional methodology involving background removal from images involves a single path having less than 6 anchor points. Intriguing designers work through such details in straight, rectangular, round, and even curved images leaving no gaps behind. Like – Mobile images, plate, ring, book, are required in this image background removal service.

Clipping Path Technique

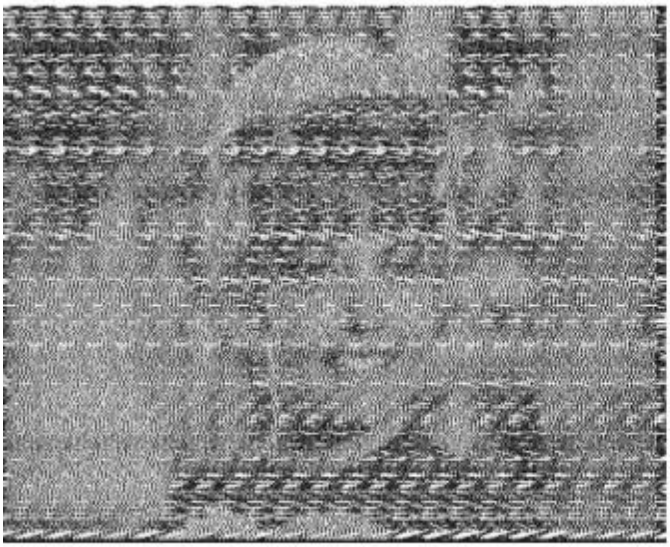
Clipping Path refers, to a close vector that is employed in such a way to cut out images and remove from them any background as well as places any desirable one. It is a remarkable tool for removing all unnecessary background from the product images and making them clear as well as unique for the viewer to have a close call for magnificent intricate detailing.



(A) Image Degraded by Blur



(B) Image Restored Using the Pseudo Inverse Filter



(C) Image in (A), Quantized with 256 Levels, after Application of the Pseudo-Inverse Filter



(D) Image in (A), Quantized with 256 Levels, after Application of the Wiener Filter

[1] Human Perception

Typical applications of these kinds of techniques are noise filtering. In somecases, the images that you get may be very very noisy. So, we have to filter those images so that the noise present in that image can be removed and the image appears much better.

In some other kind of applications, we may have to enhance certain characteristics of the image. So, the different kind of applications under this category: one is the contrast enhancement. So, sometimes the image may be very poor contrast and we have to enhance the contrast of that image so that it is better visually.

In some other cases, the image may be blurred. This blurring may occur because of various reasons. May be, the camera setting is not proper or the lens is not focused properly. So, that leads to one kind of blurring. The other kind of blurring can be if we take a picture from a moving platform; say for example, from a moving car or from a moving train. In that case also you might have absorbedthat the image that you get is not a clear image. But it is a blurred image.

The other kind of application is remote sensing. In remote sensing, the types of images are taken from a satellite.
Different examples under these different categories

Example 1

The first image that is shown in Fig. 1 is a noisy image and this kind of image is quite common on a TV screen. Now, find that the digital image processing techniques can be used to filter these images and the filtered image is shown on the right hand side and we find that the filtered image looks much better than the noisy image that have shown on the left side.

On the left hand side, a low contrast image is present but in this case it is a color image and none of us will like to have a color image of this form. On the other hand, on the right hand side, we have the same image which is enhanced by using the digital image processing techniques and the enhanced image is much better than the low contrast image.

Some cases, the image may be blurred. In Fig 3 on the top row, the left side, image is blurred and in this case, the blurring has occurred because of the defocused lens. When the image was taken, the lens was not focused properly.

On the top row, the right hand side another kind of blurring occurs if you take a picture from a moving platform, may be from a moving train or from a moving car. This kind of blurring is mostly the motion blurring

The third image on the bottom row, it shows the processed image where by processing these different blurred images, we have been able to improve the quality of the image.

Other major application of digital image processing techniques is in the area of medicine. Here, it shows CT scan image and the image is used to determine the

location of a tumor.

The left hand image is the original CT scan image and the middle and the image on the right hand side, there are the processed images. So, here in the processed image, the region of yellow and red that tell that the presence of a tumor in the brain.

UNIT-3(Image Restoration) Image Restoration

Lecture-2

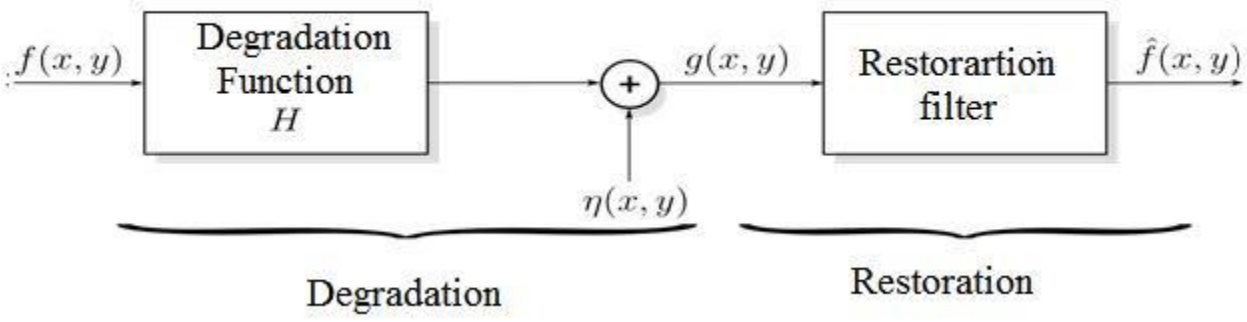
Image Restoration

In case of image enhancement or simple noise filtering, we do not make use of any of the degradation model or we do not bother about what is the process which is degrading the image.

Whereas in case of image restoration, we will talk about the degradation model, we will try to estimate the model that has degraded the image and using that model; we apply the inverse process and try to restore the image.

So, in case of image restoration, the image degradation model is very very important. So, we have to find out what is the phenomena or what is the model which has degraded the image and once that model, the degradation model is known; then we have to apply the inverse process to recover or restore the desired image.

Image degradation model



So here, we assume that our input image is image f (x, y). It is a 2 dimensional function we assume that, the input image f (x, y) is degraded by a degradation function H. So, we will put it like this that we have a degradation function H which operates on the input image f (x, y).Then, the output of this degradation function is added to an additive noise. So here, we add a noise term which we represent by say eta (x, y) which is added to the degradation output and this finally gives us the output image g (x, y).

So, this g (x, y) is the degraded image which we want to recover. So, from this g(x, y), we want to recover the input image, the original input image f (x, y) using the image restoration techniques.

So, for recovering this f (x, y), we have to perform some filtering operation. These filters, are actually derived using the knowledge of the degradation

function that is H and output of the filters is our restored image as f that (x, y) and we put it as f hat (x, y) because in most of the cases, we are unable to restore theimage exactly.

That means it is very difficult to get the exact image f (x, y) we can get an approximation of the original image f (x, y). So, that is this reconstructed image f hat (x, y) which is an approximation of the original image f (x, y).

This particular operation that is the conversion from f (x, y) to g (x, y), this can be represented in special domain as g (x, y) is equal to h (x, y) convolution with f (x, y) plus the noise eta (x, y).

The corresponding operation in frequency domain will be represented by G (u, v) is equal to H (u, v) into F (u, v) plus N (u, v) where H (u, v) is the Fouriertransformation of H (x, y), F (u, v) is the Fourier transformation of the input image f (x, y), N (u, v) is the Fourier transform of the additive noise eta (x, y) and G (u, v) is the Fourier transform of the degraded image G (x, y).

Now here, if we assume or for the time being if we neglect the term eta (x, y) or we said eta (x, y) equal to 0 for the time being for simplicity of our analysis, then what we get is g (x, y) is equal to H in f (x, y) and as we said that here this H, we assume that this is the degradation operator.

UNIT-3(Image Restoration)

Noise Model

Lecture-3

Noise tells unwanted information in digital images. Noise produces undesirable effects such as artifacts, unrealistic edges, unseen lines, corners, blurred objects and disturbs background scenes. To reduce these undesirable effects, prior learning of noise models is essential for further processing.

Noise in images:

Image noise is random variation of brightness or color information in the images captured. It is degradation in image signal caused by external sources.Images containing multiplicative noise have the characteristic that the brighter the area the noisier it. But mostly it is additive. We can model a noisy image as

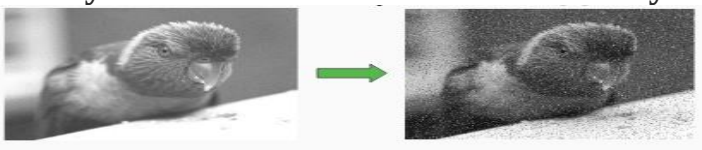


Fig.2 Addition of random noise in an image

A(x,y) = H(x,y) + B(x,y) **Where, A(x,y)= function of noisy image, H(x,y)= function of image noise , B(x,y)= function of original image.**

Sources of Image noise:

- **While image being sent electronically from one place to another.**
- **Sensor heat while clicking an image.**
- **With varying ISO Factor which varies with the capacity of camera**
- **to absorb light.**

Types of Image noise:

There are different types of image noise. They can typically be divided

into 3 types.

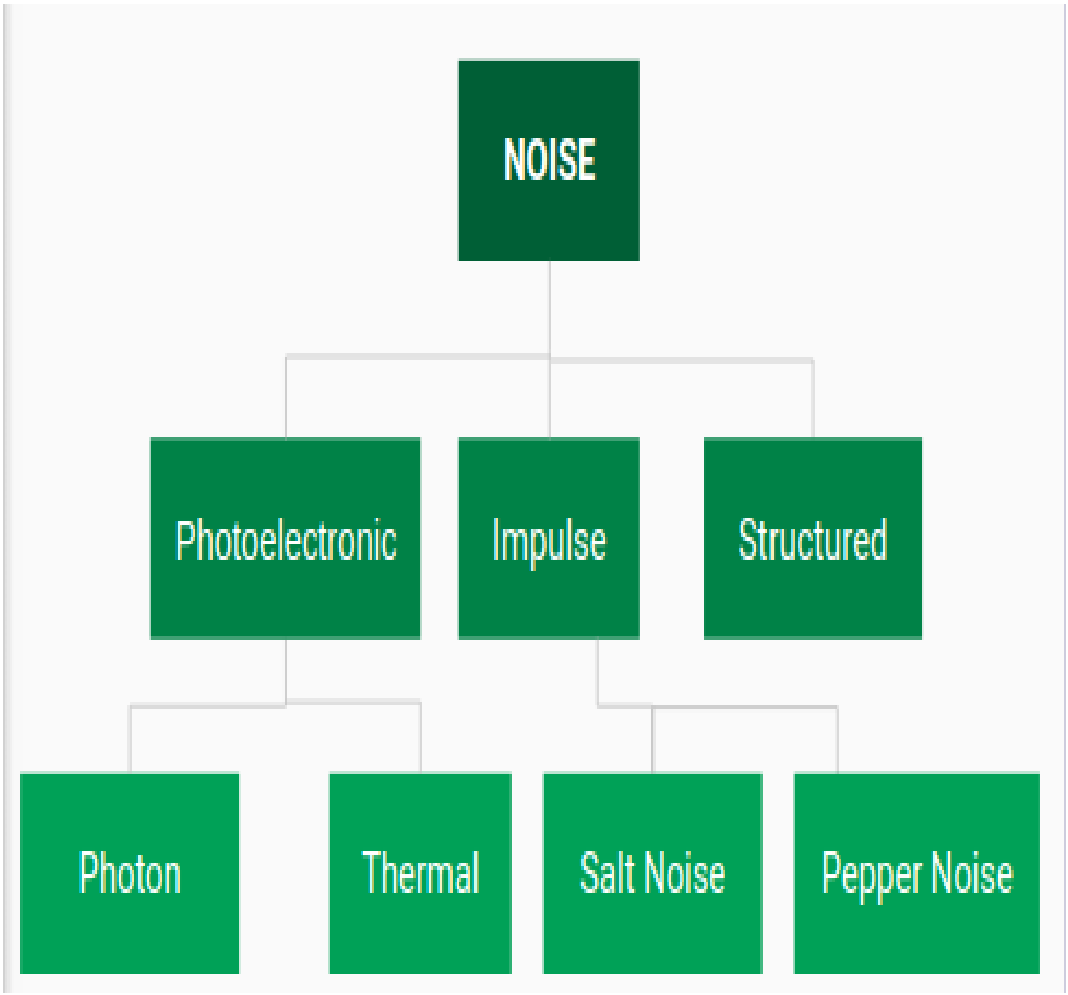


Fig. classification of noise

1. **Gaussian Noise:**

Gaussian Noise is a statistical noise having a probability density function equal to normal distribution, also known as Gaussian Distribution. Random Gaussian function is added to Image function to generate this noise. It is also called as electronic noise because it arises in amplifiers or detectors.

Source: thermal vibration of atoms and discrete nature of radiation of warm objects.



Fig.4 Plot of Probability Distribution Function

The side image is a bell shaped probability distribution function which have mean 0 and standard deviation(sigma) 1.

1.2 Effect of Standard Deviation(sigma) on Gaussian noise:

Fig.5 Effect of Sigma on Gaussian Noise

The magnitude of Gaussian Noise depends on the Standard Deviation(sigma). Noise Magnitude is directly proportional to the sigma value.

2. Impulse Noise:

Impulse Function: In the discrete world impulse function on a vale of 1 at a single location and In continuous world impulse function is an idealised function having unit area.



Fig.6 Impulse function in discrete world and continuous world

2.1 Types of Impulse Noise:

There are three types of impulse noises. Salt Noise, Pepper Noise, Salt and Pepper Noise.Salt Noise: Salt noise is added to an image by addition of random bright (with 255 pixel value) all over the image.Pepper Noise: Salt noise is added to an image by addition of random dark (with 0 pixel value) all over the image.

Salt and Pepper Noise: Salt and Pepper noise is added to an image by addition of both random bright (with 255 pixel value) and random dark (with 0 pixel value)

all over the image.This model is also known as data drop noise because statistically it drop the original data values [5]. Source: Malfunctioning of camera's sensor cell.

3. Poisson Noise:

The appearance of this noise is seen due to the statistical nature of electromagnetic waves such as x-rays, visible lights and gamma rays. The x-ray and gamma ray sources emitted number of photons per unit time. These rays are injected in patient's body from its source, in medical x rays and gamma rays imaging systems. These sources are having random fluctuation of photons. Result gathered image has spatial and temporal randomness. This noise is also called as quantum (photon) noise or shot noise.

UNIT-3(Image Restoration)Noise Filter

Lecture-4

Noise filtering in Digital Image Processing

Noise is always presents in digital images during image acquisition, coding, transmission, and processing steps.



In our last article named "Noise in Digital Image Processing" we had promised to get back to you with another article on filtering techniques and filters. So here we are with yet another article of our series "Image Vision" on Noise Filtering.

Noise is always presents in digital images during image acquisition, coding, transmission, and processing steps. It is very difficult to remove noise from the digital images without the prior knowledge of filtering techniques. In this article, a brief overview of various noise filtering techniques. These filters can be selected by analysis of the noise behaviour. In this way, a complete and quantitative analysis of noise and their best suited filters will be presented over here.

Filtering image data is a standard process used in almost every image processing system. Filters are used for this purpose. They remove noise from images by preserving the details of the same. The choice of filter depends on the filter behaviour and type of data.

Filtering Techniques:

We all know that, noise is abrupt change in pixel values in an image. So when it comes to filtering of images, the first intuition that comes is to replace the value of each pixel with average of pixel around it. This process smooths the image. For this we consider two assumptions.

Assumption:

1. The true value of pixels are similar to true value of pixels nearby
2. The noise is added to each pixel independently.

Let's first consider 1-dimensional function before going into 2-dimensional image.

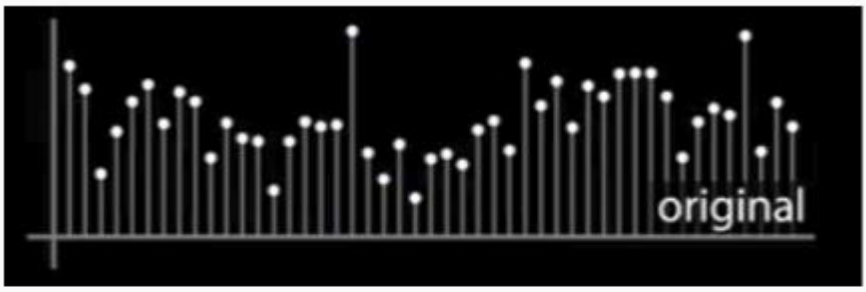


Fig.1 Original image signal in 1 Dimension with noise
Source: Udacity

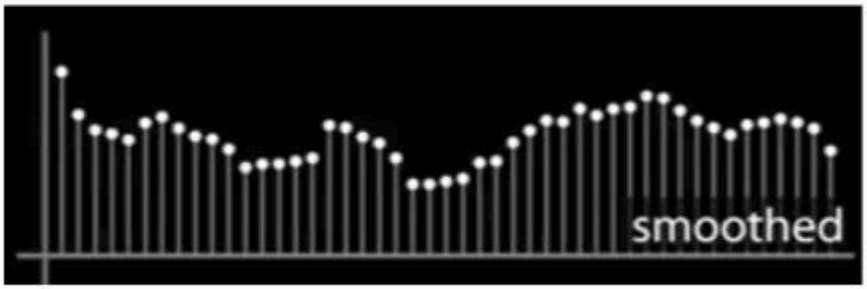


Fig.2 Image signal in 1 Dimension after averaging
Source: Udacity

In the above image of original function(fig-1), if we will consider each circle as pixel values, then the smoothed function(fig-2) is the result of averaging the side by pixel values of each pixel.

1. Filtering with weighted moving average uniform weight:

Instead of just thinking about averaging the local pixel, which is resulting in some loss of data, we consider a set of local pixel and assign them as uniform weights. Here we assume that noise is added to each pixel independently. According to this noise amount, we assign weights to different pixels.

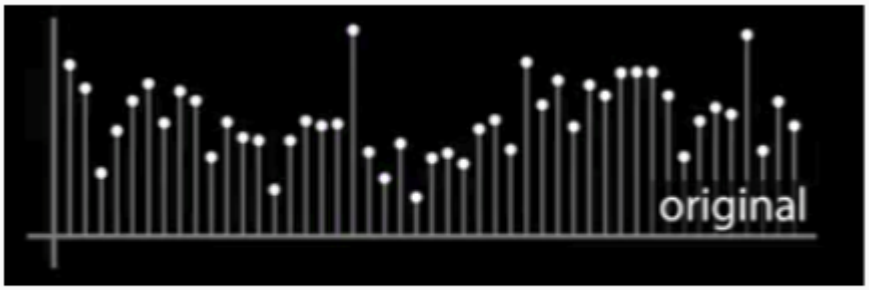


Fig.3 Original image signal in 1 Dimension with noise
Source: Udacity

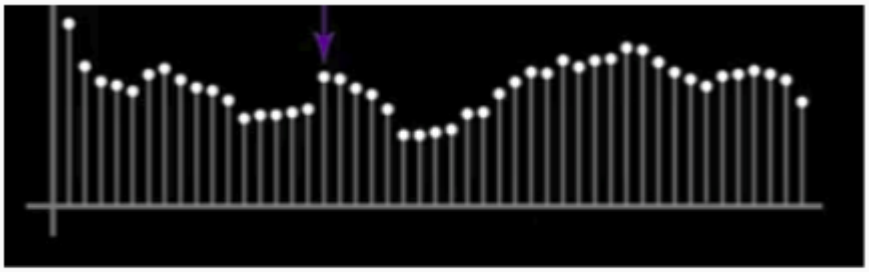


Fig.4 Image signal in 1 Dimension after uniform weighted moving averaging
Source: Udacity

2. Filtering with weighted moving average non-uniform weight

Previously we took the assumption that the true value of pixels are similar to true value of pixels nearby. But it is not always true. So for higher accuracy we assign the nearby pixels with greater weight then the pixels that are far away. This smooths the image and preserves the image information with less amount of data loss.

3. Weighted moving average in 2-dimensional image

Thinking of image as a 2-dimensional matrix, we slide a small window(the red square in fig. 5) over the whole image to replace each pixel with the average of nearby pixels. This small window is otherwise known as mask or kernel.

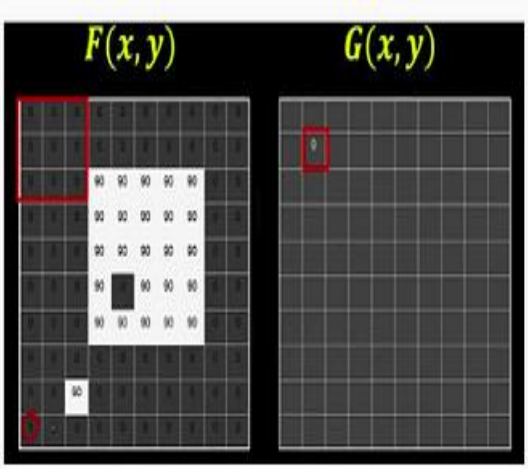


Fig.5 First step for calculating weighted moving averaging in a 2D image matrix
Source: Udacity

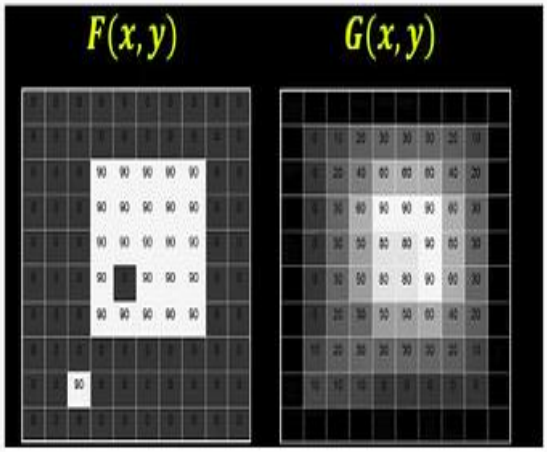


Fig.6 Last step for calculating weighted moving averaging in a 2D image matrix
Source: Udacity

3

The process used in filtering with uniform weights is also called correlation or correlation filtering.

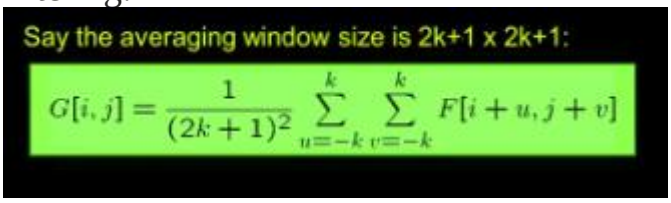


Fig.7 Correlation function for uniform weights.

In correlation filtering with non-uniform weight, an function is used as non-uniform weights which is also called mask or kernel (function of the pixel values of the small sliding window) . The process used in it is called cross-correlation.

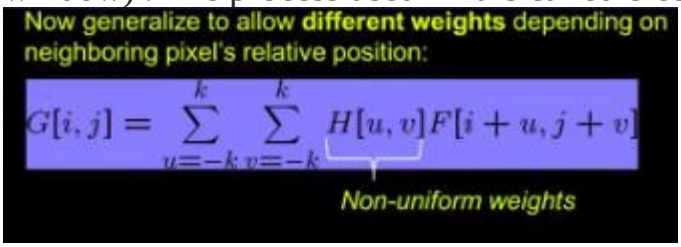


Fig.8 Correlation function for non-uniform weights

Noise Filter

Lecture-5Types of Image noise filters:

There are different types of image noise filters. They can typically be divided into 2 types.

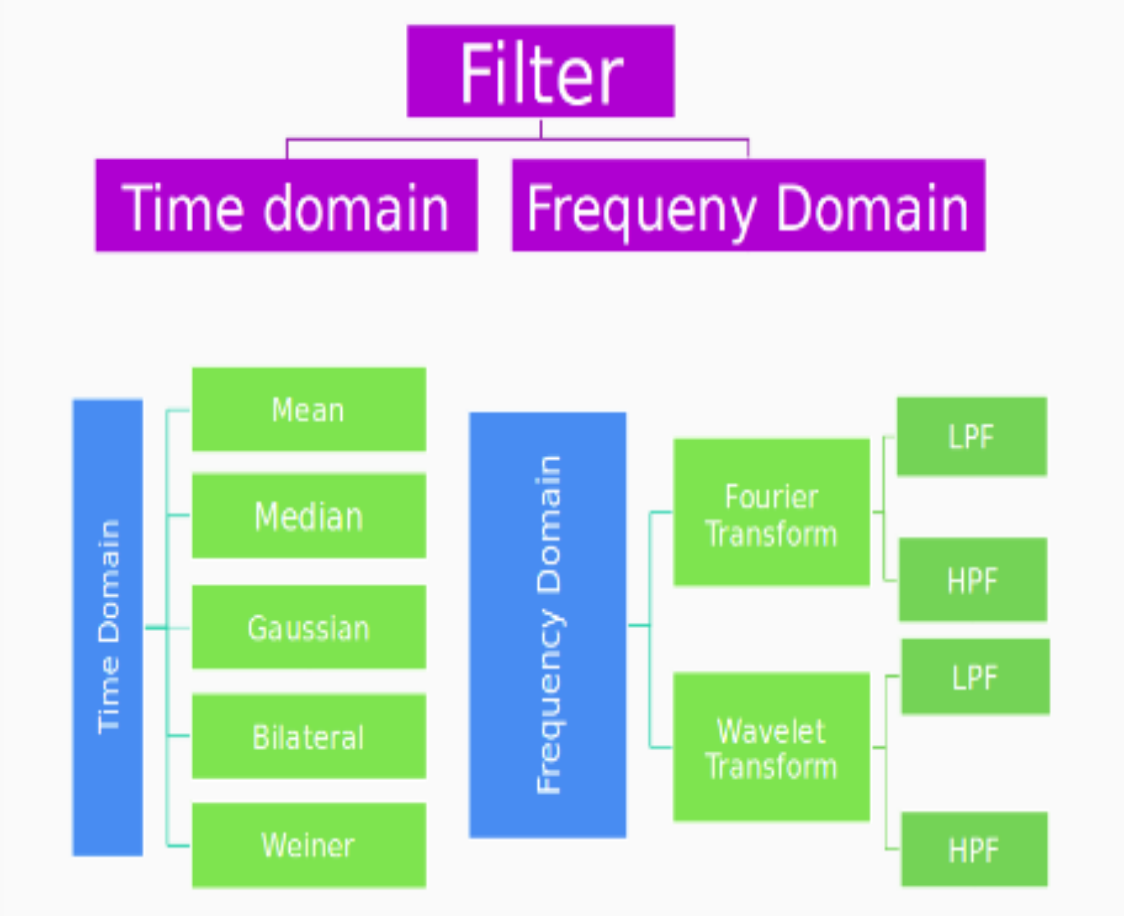


Fig. Classification of Filters

Though there are many types of filters, for this article we will consider 4 filters which are mostly used in image processing.

1. Gaussian Filter:

In image processing, a Gaussian blur (also known as Gaussian smoothing) is the result of blurring an image by a Gaussian function (named after mathematician and scientist Carl Friedrich Gauss). It is a widely used effect in graphics software, typically to reduce image noise and reduce detail.

1.1 Implementation of Gaussian Filter with OpenCV and Python:

(Filtering Gaussian Noise)

2. Mean Filter:

Mean filter is a simple sliding window that replace the center value with the average of all pixel values in the window. The window or kernel is usually a square but it can be of any shape.



Fig.10 Calculating Mean by sliding window

3. Median Filter:

Mean filter is a simple sliding window that replace the center value with the Median of all pixel values in the window. The window or kernel is usually a square but it can be of any shape.



Fig.11 Calculating Median by sliding window
(Filtering Salt and Pepper noise)

4. Bilateral Filter

Bilateral filter uses Gaussian Filter but it has one more multiplicative component which is a function of pixel intensity difference. It ensures that only pixel intensity similar to that of the central pixel is included in computing the blurred intensity value. This filter preserves edges.

Analysis of best suited filters for noises:

From the implementation of both noise and filters, we analysed best suited filters for different image noises.

NOISE	BEST SUITED FILTERS
Salt and Pepper	Median
Poisson	Mean
Gaussian	Gaussian/BilAtERAL
Speckle	Weiner

UNIT-3(ImageRestoration)

Degradation Function

Lecture-6

Image degradation is the loss of image quality for a variety of reasons. When there is image deterioration, the quality of the image is greatly diminished and becomes hazy. Image restoration is the process of enhancing or upgrading an image's quality with the aid of picture editing software.

- Image degradation is the loss of image quality for a variety of reasons. When there is image deterioration, the quality of the image is greatly diminished and becomes hazy.
- Image restoration is the process of enhancing or upgrading an image's quality with the aid of picture editing software.

Model restoration is all about how to"make up for" or "undo" imperfections that diminish an image. Shutter speed, noise, and camera focus issues are just a few examples of how things might get worse. In situations when motion blur occurs, it is feasible to predict the blurring function quite accurately and "undo" the blurring to regain the original image. The best we can expect when noise distorts an image is to make up for the damage it caused. Here are a few techniques for picture restoration utilized in the realm of image processing.

- Reverse Filter. This approach involves viewing an image degrader while assuming a well-known blurring function. You'll observe that restoration works well when there is no noise and less well when there is.
- Weiner Filters. In this section, you put wiener filtering to use to restore images, giving you the best possible balance between inverse and de-noising filtering. You'll observe that the outcome is generally superior to that of inverse filtering done directly.
- Wavelet Restoration. To recover the image, you use three wavelet-based methods.
- Blind Deconvolution. You make no assumptions about the image with this technique. You are completely ignorant in regards to the blurring feature or the added noiset. You'll discover how challenging it is to restore an image when you don't know anything about it.

UNIT-3(Image Restoration)

Lecture-7

Inverse Filter

If we know of or can create a good model of the blurring function that corrupted an image, the quickest and easiest way to restore that is by inverse filtering. Unfortunately, since the inverse filter is a form of high pass filer, inverse filtering responds very badly to any noise that is present in the image because noise tends to be high frequency. In this section, we explore two methods of inverse filtering - a thresholding method and an iterative method.

Method 1: Thresholding

Theory

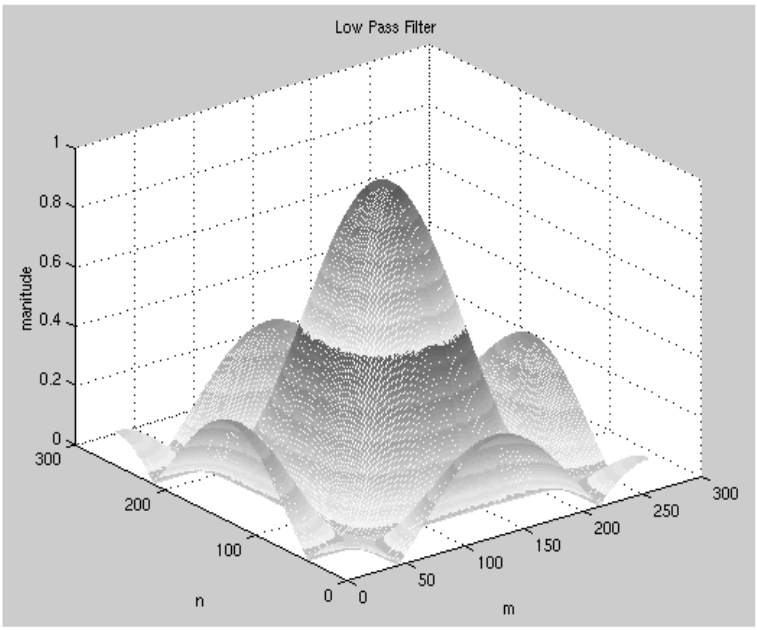
We can model a blurred image by

$$g(n_1, n_2) = f(n_1, n_2) * *b(n_1, n_2)$$

where *f* is the original image, *b* is some kind of a low pass filter and *g* is our blurred image. So to get back the original image, we would just have to convolve our blurred function with some kind of a high pass filter

$$f(n_1, n_2) = g(n_1, n_2) * *h(n_1, n_2)$$

But how do we find h? If we take the DFT of *b* so that **B**=DFT2(*b*), we would get something that looks like this



Problem in inverse filter is that noise is unknown. It can not be remove properly that problem is solved in wiener filter.

The Wiener filter is the MSE-optimal stationary linear filter for images degraded by additive noise and blurring.

If noise is zero, wiener filter reduces to inverse filter.

Noise power spectrum is constant when dealing with white noise and In wiener filter K is adjusted to get the best result.

Drawback:

Must know (or approximate) the power spectra of undegraded image and noise.

Unit-3 (Image Restoration)

Lecture-8

Homomorphism Filtering

Homomorphic filtering is a frequency-domain technique that aims at a simultaneous increase in contrast and dynamic range compression. It is mainly utilized for non-uniformly illuminated images in medical, sonar images etc. [8] for edge enhancement that makes the image details clear to the observer.

The image can be characterized by two components,

(1) the amount of source illumination incident

(2) amount of illumination reflected by the object.

These are called the illumination and reflectance components of the image.

The illumination-reflectance model can be used to develop a frequency domain procedure for improving the appearance of an image by simultaneous gray-level range compression and contrast enhancement. In this application, the key to the approach is the separation of the illumination and the reflectance components.

Homomorphic Filtering

An image as a function can be expressed as the product of illumination and reflectance components as follows:

$$F(x,y) = I(x,y) * R(x,y) \quad (1)$$

Equation (1) cannot be used directly to operate separately on the frequency components of illumination and reflectance because the Fourier transform of the product of two functions is not separable. Instead the function can be represented as a logarithmic function wherein the product of the Fourier

transform can be represented as the sum of the illumination and reflectance components.

Homomorphic filtering:

The illumination-reflectance model can be used to develop a frequency domain procedure for improving the appearance of an image by simultaneous gray-level range compression and contrast enhancement. An image f(x, y) can be expressed as the product of illumination and reflectance components:

$$f(x, y) = i(x, y)r(x, y).$$

Equation above cannot be used directly to operate separately on the frequency components of illumination and reflectance because the Fourier transform of the product of two functions is not separable; in other words,

$$\Im\{f(x, y)\} \neq \Im\{i(x, y)\}\Im\{r(x, y)\}.$$

Suppose, however, that we define

$$\begin{aligned} z(x, y) &= \ln f(x, y) \\ &= \ln i(x, y) + \ln r(x, y). \end{aligned}$$

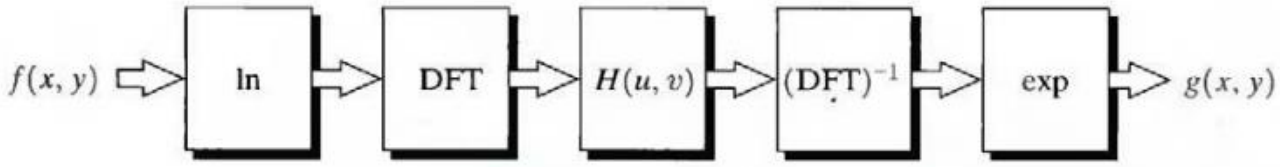
$$s(x, y) = i'(x, y) + r'(x, y).$$

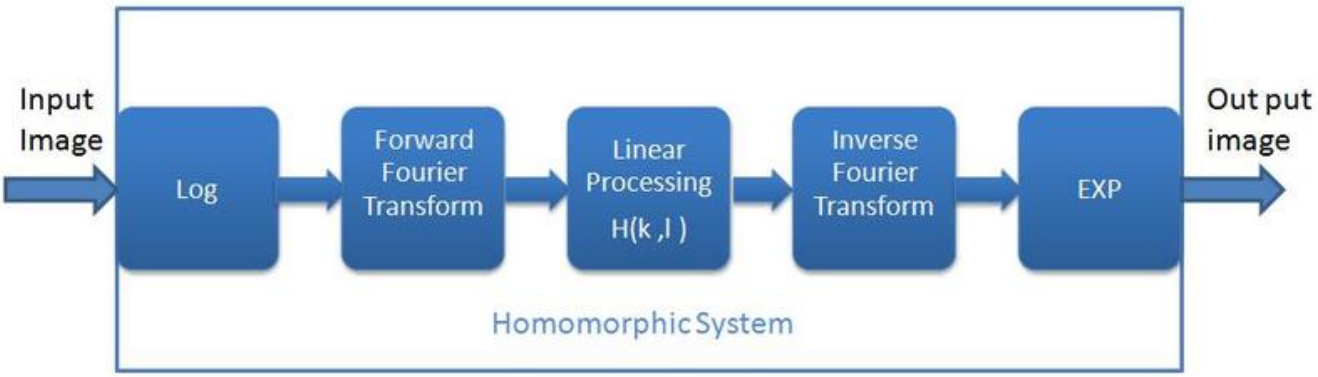
Finally, as z (x, y) was formed by taking the logarithm of the original image f (x, y), the inverse (exponential) operation yields the desired enhanced image, denoted by g(x, y); that is,

$$\begin{aligned} g(x, y) &= e^{s(x,y)} \\ &= e^{i'(x,y)} \cdot e^{r'(x,y)} \\ &= i_0(x, y)r_0(x, y) \end{aligned}$$

where

$$i_0(x, y) = e^{i'(x,y)}$$





Examples

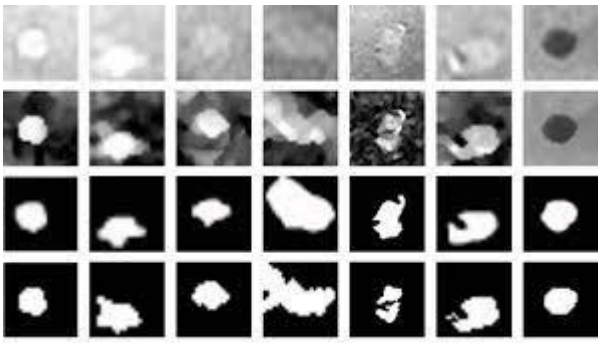


Fig showing homorphic filtering